Gradients & the Geometry of Surfaces

The arrow that points straight uphill

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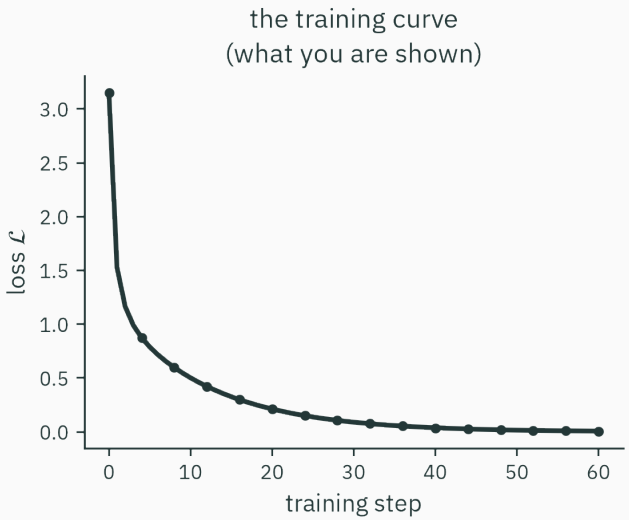
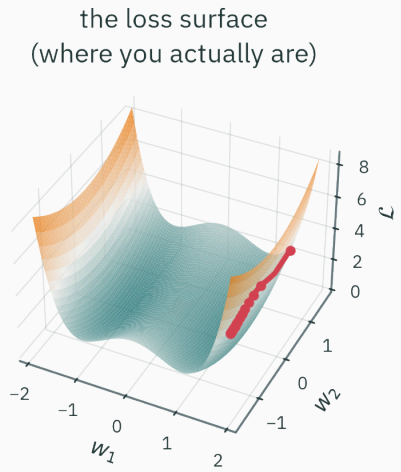
Mathematical Foundations for AI · IIT Gandhinagar

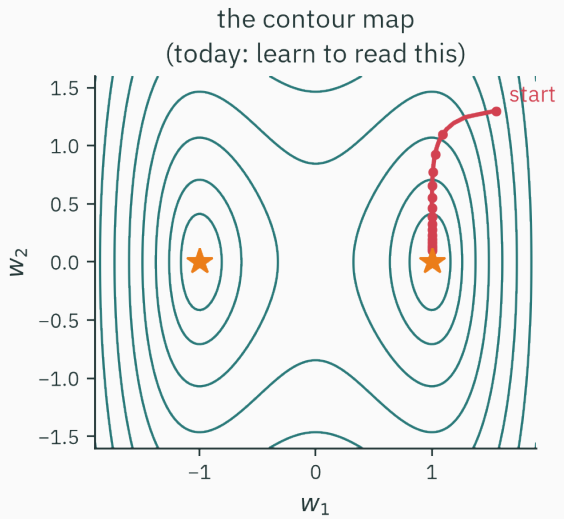
The gradient of a two-parameter loss

Every training run hides a landscape

V VISUAL







You will stare at thousands of the left plot. It is an **altimeter log**: height vs time of a walk. The walk itself happens on the **middle** object — and hikers navigate it with the **right** one.

Training is a walk you cannot watch

- A real model has **millions** of weights — its loss surface lives in $\mathbb{R}^{10^6}$, unplottable forever.
- The core local mechanisms — slopes, contours, and steepest directions — are already visible with **two** weights.
- So today is deliberately flat-footed: $f(x, y)$, drawn every way we can.

Training = walking downhill on a loss surface. The training curve is only the altimeter — today we learn to read the terrain.

Today’s one idea

The gradient collects every partial sensitivity into one vector. At a regular point it gives steepest ascent and is perpendicular to the local contour.

Three questions, in order:

1. How do we even **draw** a function of two variables? → surfaces & contour maps
2. How do we **differentiate** it? → partial derivatives, bundled into the gradient
3. Which direction is **steepest**, and why? → the directional derivative ⭐

Learning outcomes

By the end of this lecture you will be able to:

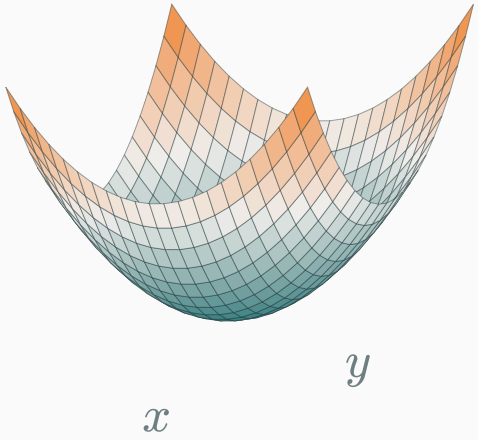
1. Draw one $f(x, y)$ two ways — **surface** and **contour map** — and translate between them.
2. Read a contour map **like a hiker**: peaks, pits, saddles, steep faces.
3. Compute **partial derivatives** by freezing variables; check them numerically.
4. Assemble the **gradient** ∇f and interpret it as an arrow on the map.
5. Predict the slope in **any** direction from just ∇f — via $\nabla f \cdot \boldsymbol{u}$.
6. Prove ⭐ that ∇f is **perpendicular to contours** and is the **steepest** direction.
7. Run the **multivariate chain rule** along branching pipelines.

Surfaces and contour maps

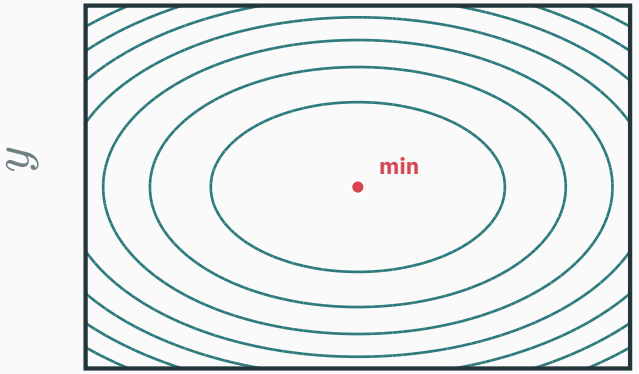
A function of two inputs is a landscape

V VISUAL

$f(x, y) = x^2 + 3y^2$ — every map point (x, y) gets one number: its **height**.



the **surface** view: height drawn as height



the **map** view: height drawn as curves

Same f, two pictures. The rest of the course lives in the right one — so let's earn it.

Your first loss landscape, built by hand

Fit the line $\hat{y} = wx + b$ to two points $(1, 2)$ and $(2, 4)$. The loss is a function of **two** variables — the parameters:

$$\mathcal{L}(w, b) = (2 - (w + b))^2 + (4 - (2w + b))^2$$

w	b	$\mathcal{L}(w, b)$
0	0	$4 + 16 = 20$
1	1	$0 + 1 = 1$
2	1	$1 + 1 = 2$
2	0	$0 + 0 = \mathbf{0}$

Each row: guess a line, square the two misses, add.

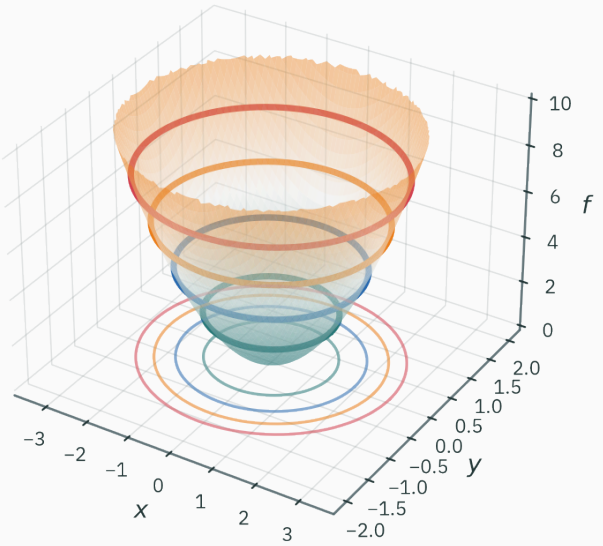
$(w, b) = (2, 0)$ fits perfectly — the data **is** $y = 2x$.

Keep this $\mathcal{L}$ in mind; it returns at the end of the lecture, with a walker on it.

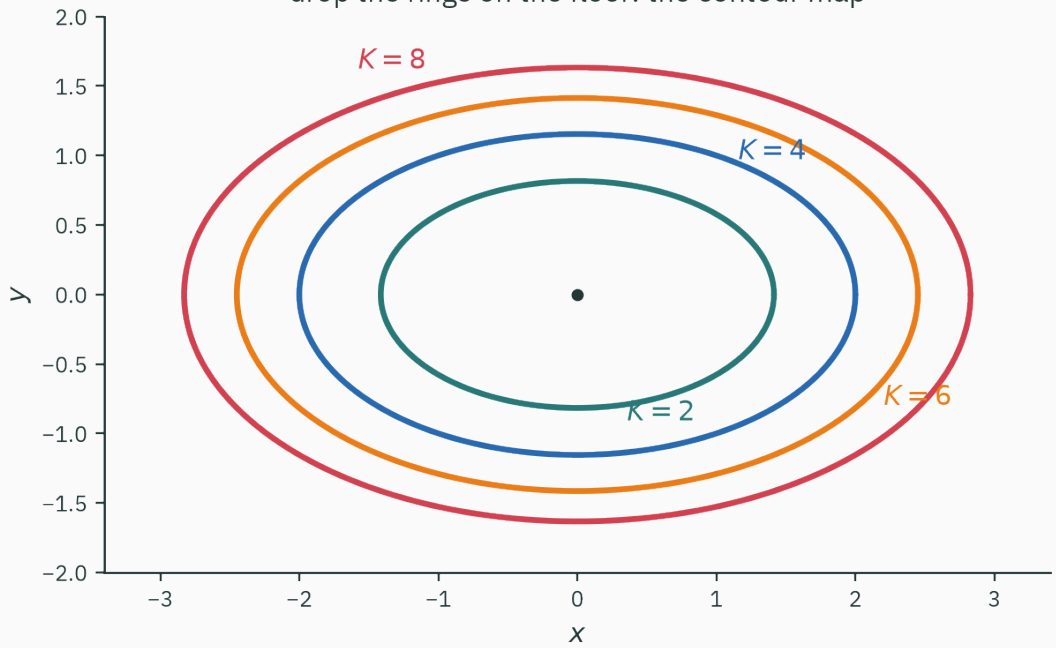
How a contour map is made

V VISUAL

slice at constant heights $f(x, y) = K$



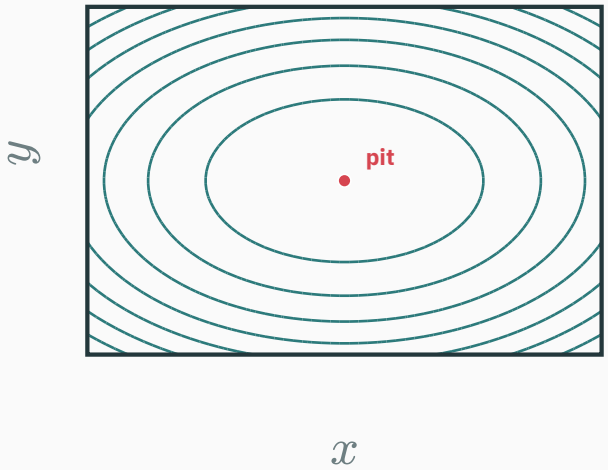
drop the rings on the floor: the contour map



Slice the surface at constant heights $f(x, y) = K$; drop each cut edge straight down. A **contour line** (or **level set**) is everything at one height.

Read the map like a hiker

- **Closed rings shrinking to a dot** — a peak or a pit lives inside.
- **Lines crowded together** — steep face: height changes fast per step.
- **Lines far apart** — gentle plain: many steps to change height.
- **Walking along a line** — you stay exactly level, by definition.



Equal height spacing between lines ⟹ line crowding = steepness. A contour map is a slope chart in disguise.

The map in code · meshgrid + contour

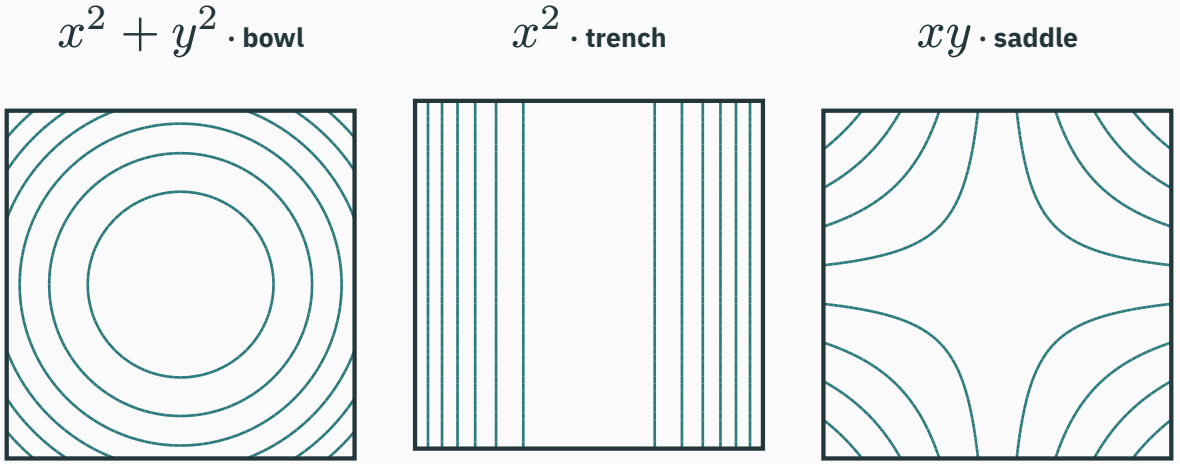
```
import numpy as np, matplotlib.pyplot as plt
x = np.linspace(-3, 3, 100)
y = np.linspace(-2, 2, 100)
X, Y = np.meshgrid(x, y)      # X[i,j], Y[i,j]: every grid pair
Z = X**2 + 3*Y**2             # f on the whole grid at once
plt.contour(X, Y, Z, levels=10)
plt.gca().set_aspect("equal")
```

meshgrid tabulates the plane; contour interpolates level curves between grid values. These five lines are the basic recipe behind many two-dimensional loss-landscape plots.

T4 walks through meshgrid slowly — including the classic confusion about which axis is which.

A gallery of terrains

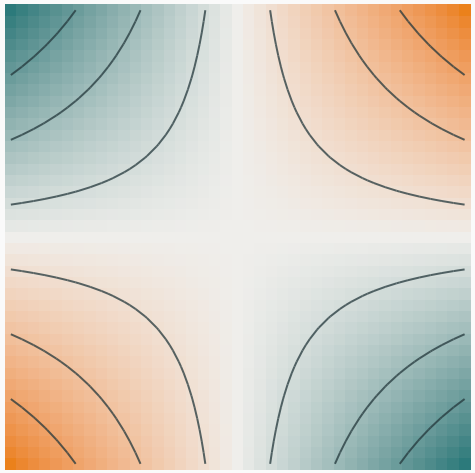
V VISUAL



- **Bowl**: rings around one pit — the ideal loss. **Trench**: f **ignores** y — a loss that ignores a parameter (a dead weight: contours parallel to its axis).
- **Saddle**: downhill in two quadrants, uphill in two — no ring closes.

The saddle deserves a second look

V VISUAL



heat view: high / low

At the center: flat — yet **neither** a peak nor a pit. Ride $x = y$ and you climb; ride $x = -y$ and you sink.

High-dimensional objectives can have saddle critical points: zero-gradient locations that are not minima. L9 introduces the Hessian, whose curvature signs distinguish the local quadratic models of bowls, saddles, and troughs.

Level sets you have already met

Contours are not just a plotting trick — they are how ML **draws decisions**:

- With the usual binary threshold 0.5, a classifier's **decision boundary** is the level set $p(x, y) = 0.5$. Different costs or thresholds move that boundary.
- Every "decision boundary" plot in ES 335 is `plt.contour(X, Y, P, levels=[0.5])` — the same five lines of code as before.

Same object, two costumes: in **parameter space**, level sets of the loss (today); in **input space**, level sets of the model's output (your next ML course).

Partial derivatives: freeze one variable

A partial derivative varies one coordinate at a time

L7′s derivative answers: nudge **the** input, how hard does the output move? Now there are **two** inputs to nudge — and they can move together, or alone.

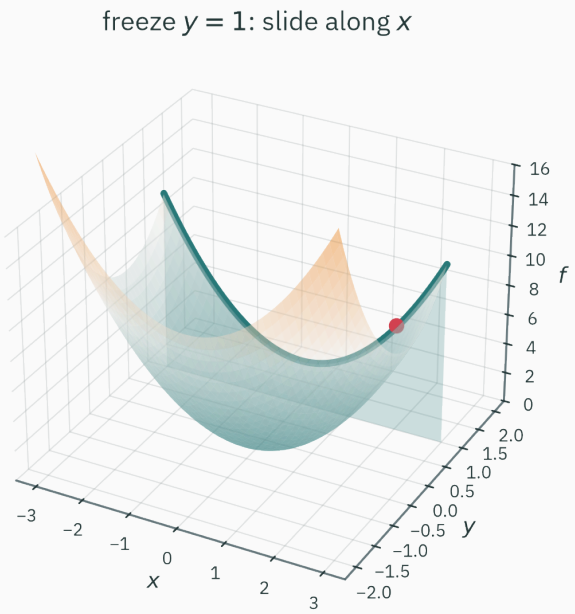
The multivariable move (you will use it for the rest of your life):

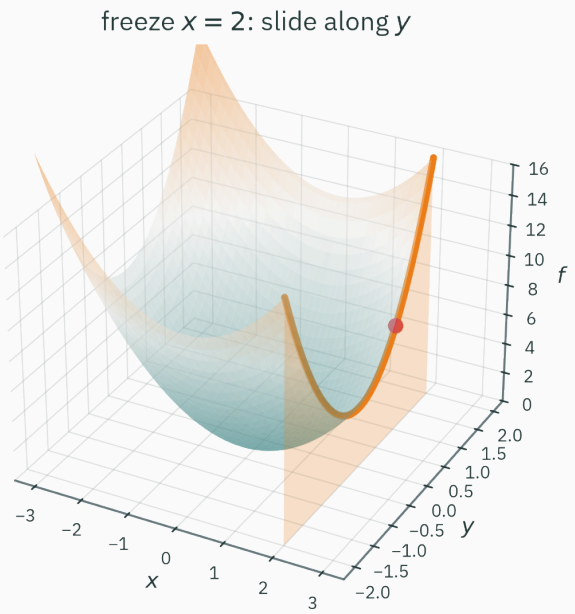
> **Freeze every variable except one. What remains is a one-variable function — and L7 already taught you everything about those.**

Freezing has a picture, a number, and a symbol. Picture first.

Freezing a variable saws the surface open

V VISUAL

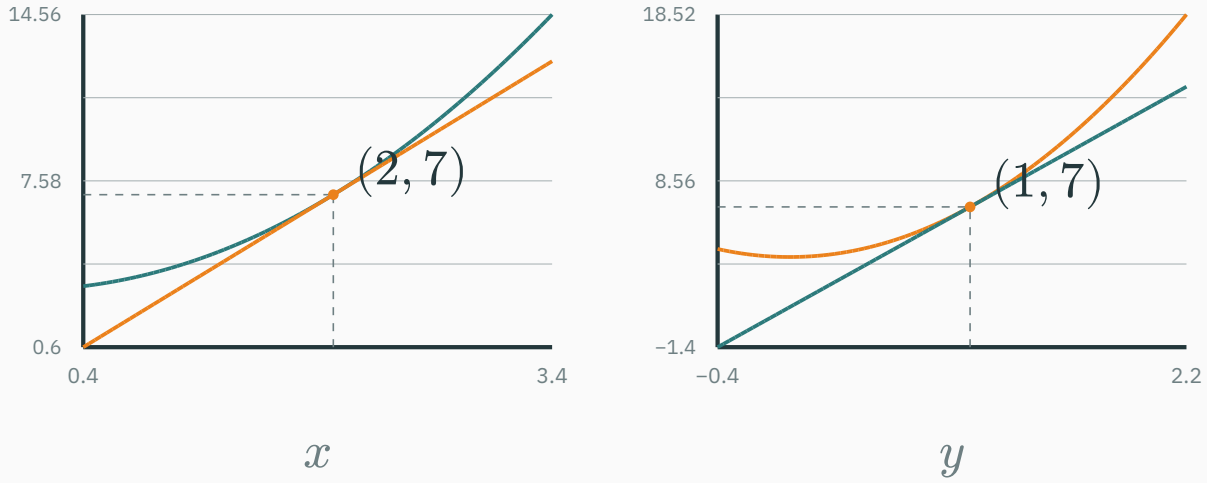




Each freeze exposes a **cut edge** through our base point $(2, 1)$, where $f = 7$ (the dot).

Each cut edge is an ordinary curve

$$g(x) = f(x, 1) = x^2 + 3 \qquad h(y) = f(2, y) = 4 + 3y^2$$



Each cut is a parabola with a **tangent line** at the base point — exactly L7′s local-line approximation. Two cuts, two slopes. Let’s measure them.

The nudge table, twice

Base point $(2, 1)$, $f = 7$. Nudge **one** coordinate, hold the other — L7′s ritual:

move	new value	response	ratio
$x : 2 \to 2.01$	$f(2.01, 1) = 7.0401$	0.0401	4.01
$x : 2 \to 2.001$	$f(2.001, 1) = 7.004001$	0.004001	4.001
$y : 1 \to 1.01$	$f(2, 1.01) = 7.0603$	0.0603	6.03
$y : 1 \to 1.001$	$f(2, 1.001) = 7.006003$	0.006003	6.003

Two stabilizing ratios: the x-ratio heads to 4, the y-ratio to $\mathbf{6}$.

At $(2, 1)$ this surface is 4-steep along x and 6-steep along y. Two sensitivities, one point.

∂ joins the notation zoo

L7 promised "∂ joins the zoo in L8". Here it is — the **partial derivative**:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

- Curly ∂ = "I nudged **one** input; every other input is frozen."
- To compute: **differentiate as usual, treating the frozen variables as constants**:

$$f = x^2 + 3y^2: \quad \frac{\partial f}{\partial x} = 2x + 0 = 2x, \quad \frac{\partial f}{\partial y} = 0 + 6y = 6y$$

At $(2, 1)$: $2x = \mathbf{4}$ ✓ and $6y = \mathbf{6}$ ✓ — exactly the two stabilized ratios.

Partials in code

```
import torch
f = lambda x, y: x**2 + 3*y**2
x0, y0 = torch.tensor(2.0), torch.tensor(1.0)

df_dx = torch.func.grad(f, argnums=0)(x0, y0)   # nudge arg 0 only
df_dy = torch.func.grad(f, argnums=1)(x0, y0)   # nudge arg 1 only
print(df_dx, df_dy)
# tensor(4.)  tensor(6.)
```

`argnums` **is** the freeze: differentiate w.r.t. argument 0, hold the rest. And the values are exact — no `0.0401`-style nudge noise. How PyTorch manages that is the whole of L10–L11.

Checkpoint: read the map like a pro

Q QUESTION

On a loss contour map, a small step in the w_2 direction crosses three contour lines; the same-length step in the w_1 direction stays between two lines. At that point:

A. $|\partial \mathcal{L}/\partial w_1| > |\partial \mathcal{L}/\partial w_2|$

B. $|\partial \mathcal{L}/\partial w_2| > |\partial \mathcal{L}/\partial w_1|$

C. the two partials are equal

D. $\nabla \mathcal{L} = \mathbf{0}$ there

Answer: read the map like a pro

A ANSWER

Correct: B — the w_2 partial is bigger

Why: Contour lines sit at equal height steps — crossing three of them means the height changed three steps' worth over one small move: a big ratio, i.e. a big $|\partial \mathcal{L}/\partial w_2|$. Staying between lines means barely any change: small $|\partial \mathcal{L}/\partial w_1|$. D is the opposite of what's described: a zero gradient point has **locally no** crossings in any direction.

The gradient: every sensitivity in one arrow

Bundle the partials

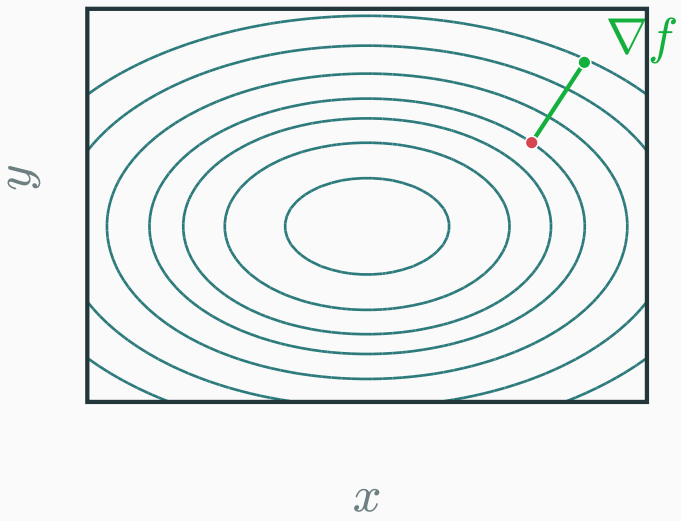
Two sensitivities at every point beg to be one object. Stack them:

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} \qquad \text{here:} \quad \nabla f = \begin{pmatrix} 2x \\ 6y \end{pmatrix}, \quad \nabla f(2, 1) = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

- ∇f (“nabla f”, or **the gradient**) is a **vector-valued** recipe: feed a point, get a vector.
- It packages **every** first-order fact about f at that point — L7′s derivative, grown up.

Honest bookkeeping: MML writes ∇f as a **row** vector (good reasons arrive with the Jacobian). We draw it as a column/arrow today; L9 settles the layout conventions.

The gradient is an arrow on the map



The pin planted at $(2, 1)$ is $\nabla f = (4, 6)$, drawn **in the input plane**.

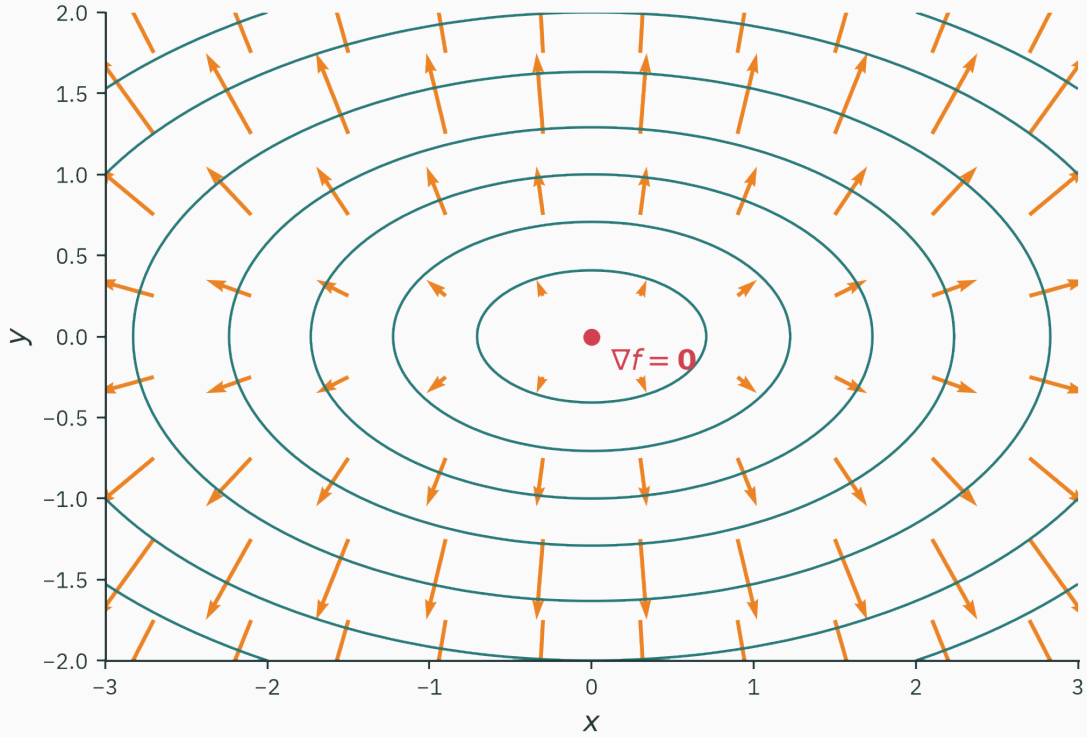
It leans more along y than x ($6 > 4$) — the surface is steeper that way.

The gradient is **not** an arrow on the mountain. It lives on the **map**, pointing along the ground — "walk that way to climb fastest."

Drop an arrow at every point: the gradient field

V VISUAL

Every arrow is ∇f planted at its base point — three regularities hide here:



What the field is telling us

1. Every arrow points **away from the pit** — uphill, toward higher contours.
2. Wherever $\nabla f \neq \mathbf{0}$ and the level set is smooth, the gradient crosses its contour at a right angle.
3. Arrows are **long where lines crowd** (steep!), and shrink to $\nabla f = \mathbf{0}$ at the flat bottom — the resting places of gradient descent (L16–L18′s obsession).

At a regular point ($\nabla f \neq 0$), the gradient is perpendicular to the contour through the point, and its norm is the maximum directional derivative.

The field in code

```
qx, qy = np.meshgrid(np.linspace(-2.7, 2.7, 10),
                     np.linspace(-1.7, 1.7, 8))
U, V = 2*qx, 6*qy                  # ∇f = (2x, 6y) on the grid
plt.contour(X, Y, Z, levels=10)
plt.quiver(qx, qy, U, V)           # arrows over the map
```

`quiver` = “a case for arrows”. Contours from the **values** of f, arrows from its **partials** — the two halves of today, overlaid.

Which way is steepest?

A hiker’s three questions

Standing at $(2, 1)$ on our surface, gradient in hand — $\nabla f = (4, 6)$:

1. If I walk **north-east** — or any direction I fancy — how fast do I climb?
2. Which direction climbs **fastest**?
3. Which directions keep me exactly **level**?

One dot product will answer all three. First, make question 1 precise — with numbers.

Walking in a chosen direction

Pick a **unit** direction $\boldsymbol{u}$, walk along $(2,1)+t\,\boldsymbol{u}$, and log altitude: $\varphi(t)=f((2,1)+t\boldsymbol{u})$. The slope $\varphi'(0)$ is the **directional derivative** $D_{\boldsymbol{u}}f$. Measure it, nudge-style:

direction $\boldsymbol{u}$	**measured slope**	**recognize it?**
$(1,0)$ — due east	4	$\partial f/\partial x$ — of course: freezing y **is** walking east
$(0,1)$ — due north	6	$\partial f/\partial y$
$(0.6,0.8)$ — north-east-ish	7.2	$0.6\cdot 4+0.8\cdot 6=7.2$ …a **blend** of the two

The diagonal slope is the partials weighted by the direction's components. That pattern is the whole theorem — now we derive it.

⭐ Step 1 · zoom in: the local plane

D DERIVATION

L7: zoom into a smooth **curve** and only a line survives. Zoom into a smooth **surface** and only a **plane** survives — tilted 4-steep along x, 6-steep along y:

$$f(2+h,1+k) \approx \underbrace{7}_{f(2,1)} + \underbrace{4}_{\partial f/\partial x}\, h + \underbrace{6}_{\partial f/\partial y}\, k$$

Why: move in two legs. East by h: the x-cut's tangent charges $4h$. Then north by k: the y-cut's tangent charges $6k$. (To first order the second leg's slope hasn't changed — that's what **smooth** buys.)

Test it at $(h,k) = (0.1, 0.05)$: plane says $7 + 0.4 + 0.3 = 7.70$; truth $f(2.1, 1.05) = 7.7175$. Off by 0.0175 — a second-order error, as L7 predicts.

⭐ Step 2 · the slope along u is a dot product

D DERIVATION

Walk the plane in direction $\boldsymbol{u} = (u_1, u_2)$: set $(h, k) = (tu_1, tu_2)$,

$$\varphi(t) \approx 7 + t\,(4u_1 + 6u_2) \quad \Rightarrow \quad \varphi'(0) = 4u_1 + 6u_2$$

That coefficient is exactly a dot product — and nothing about $(4, 6)$ was special:

$$D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u}$$

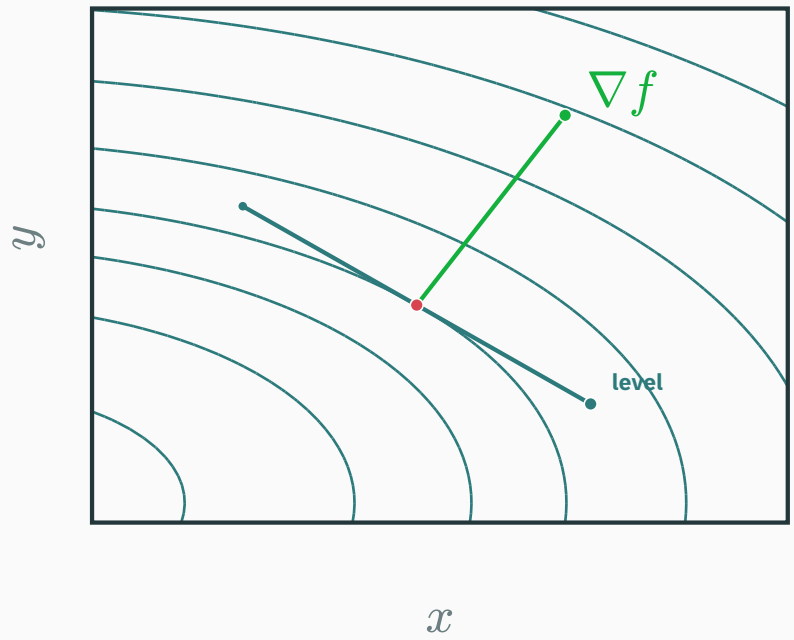
Check against the measured table: $\boldsymbol{u} = (0.6, 0.8)$: $(4)(0.6) + (6)(0.8) = 7.2$ ✓. Axes: $(1, 0) \to 4$ ✓, $(0, 1) \to 6$ ✓.

One vector, ∇f, predicts the slope in every direction. A first-order optimizer uses this information to choose a local descent direction.

The right angle, up close

V VISUAL

Zoom onto $(2, 1)$: the exact ∇f (green pin) vs the level curve $f = 7$ and its direction (teal chord):



The claim: $90°$ at every **regular point** of a smooth level set, where $\nabla f \neq \mathbf{0}$. Picture argument, then algebra.

⭐ Step 3 · level walking ⟹ perpendicular

D DERIVATION

The picture argument first. Walking **along a contour**, your altitude is pinned — the level set is the set of one height. No change, no slope:

$$D_{\boldsymbol{u}} f = 0 \quad \text{whenever } \boldsymbol{u} \text{ hugs the contour}$$

Now the algebra finishes it in one line. $D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u} = 0$, and L3 taught us exactly what a zero dot product **means**:

> $\nabla f \cdot \boldsymbol{u} = 0 \Leftrightarrow \nabla f \perp \boldsymbol{u}$**: the gradient is perpendicular to the level set through its point.**

Numeric witness at $(2, 1)$: the contour-hugging direction is $\propto (3, -2)$, and $(4, 6) \cdot (3, -2) = 12 - 12 = 0$ ✓. Claim 2 of the quiver plot — proved.

⭐ Step 4 · steepest ascent, from Cauchy–Schwarz

D DERIVATION

Among all unit $\boldsymbol{u}$, which maximizes $D_{\boldsymbol{u}} f$? L3′s second reading of the dot product:

$$D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u} = \|\nabla f\| \underbrace{\|\boldsymbol{u}\|}_{=1} \cos\theta = \|\nabla f\| \cos\theta$$

$\cos\theta \le 1$, with equality only at $\theta = 0$ — this is L3′s Cauchy–Schwarz, “bought for nothing”:

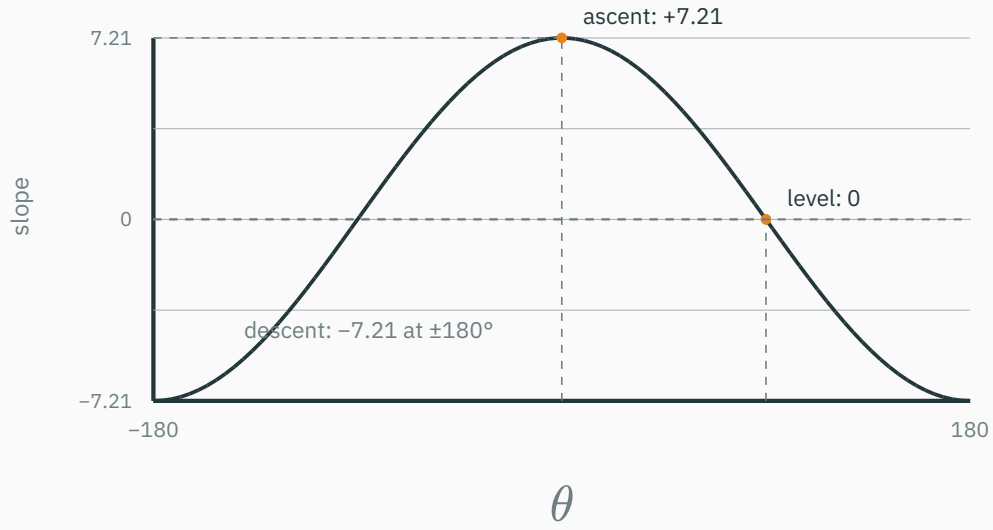
- **Steepest ascent**: $\boldsymbol{u} = \nabla f / \|\nabla f\|$, slope $\|\nabla f\| = \sqrt{4^2 + 6^2} = \sqrt{52} \approx 7.21$
- **Steepest descent**: $-\nabla f$, slope -7.21 — the direction training will take (L17)

Our hand-picked $(0.6, 0.8)$ scored 7.2 of a possible 7.21 — it sits just $3°$ off the true champion $(4, 6)/\sqrt{52} \approx (0.55, 0.83)$. Close only counts because $\cos\theta$ is flat near $\theta = 0$.

The slope compass

V VISUAL

Slope at $(2, 1)$ vs the angle θ (in degrees) between your step and ∇f — a pure cosine:



The directional derivative is maximal along ∇f, zero along contour directions, and minimal along $-\nabla f$.

Checkpoint: the level walk

Q QUESTION

At some point on a surface, $\nabla f = (3, 4)$. You step in the unit direction $u = (4, -3)/5$. The instantaneous rate of change of f is:

A. 5

B. 0

C. -5

D. 7

Answer: the level walk

A ANSWER

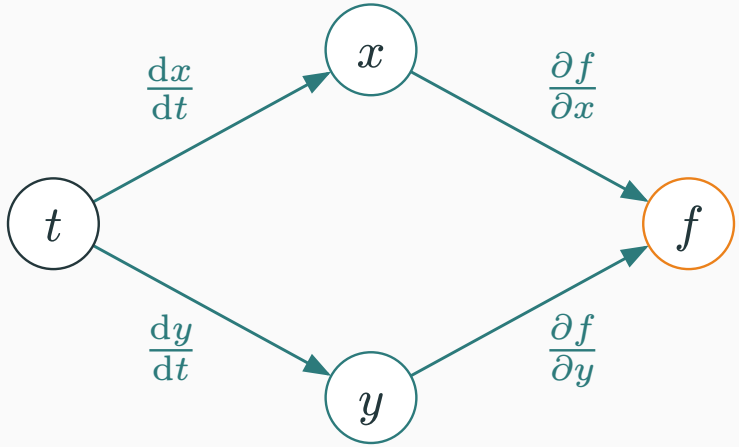
Correct: B — 0 — you walked along the contour

Why: $D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u} = (3 \cdot 4 + 4 \cdot (-3))/5 = 0$: this $\boldsymbol{u}$ is perpendicular to ∇f, so it is a level direction. A is the rate along $\nabla f / \|\nabla f\|$; C is the rate along the negative-gradient direction; D incorrectly adds components.

The chain rule grows branches

Composition, now with two routes

L7′s chain rule ran along a single pipeline. In two variables the pipeline **branches** — one clock t drives both coordinates of a moving point:



$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

multiply along each route, **add** the routes

Numbers first: a drone over the bowl

A drone flies the path $x = t$, $y = t^2$ over our surface $f = x^2 + 3y^2$. How fast is the terrain rising under it at $t = 1$ (position $(1, 1)$)?

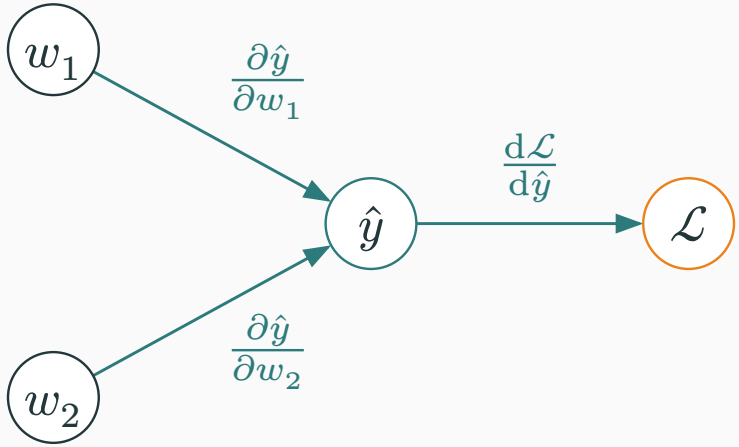
Route by route. $\partial f/\partial x = 2$, $\partial f/\partial y = 6$ at $(1, 1)$; $\mathrm{d}x/\,\mathrm{d}t = 1$, $\mathrm{d}y/\,\mathrm{d}t = 2t = 2$:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = 2 \cdot 1 + 6 \cdot 2 = 14$$

Sanity check, no chain rule. Substitute first: $f(t, t^2) = t^2 + 3t^4$, so $\mathrm{d}f/\,\mathrm{d}t = 2t + 12t^3 = 14$ at $t = 1$ ✓.

Why **add**? Nudge t by 0.01: the x-route delivers $2 \times 0.01 = 0.02$ of height, the y-route $6 \times 0.02 = 0.12$. Both arrive: $\Delta f \approx 0.14$. (Truth: $f(1.01, 1.0201) = 4.1419$, i.e. $\Delta f = 0.1419$.)

The chain ML actually runs



- Forward: weights make a prediction, prediction makes a loss — routes **into** $\mathcal{L}$.
- Training needs $\partial\mathcal{L}/\partial w_i$ for **every** weight — the gradient $\nabla_{\boldsymbol{w}}\mathcal{L}$, a million partials.

Multiply along routes, add the routes — at industrial scale, with every route shared. Doing that bookkeeping without waste is backpropagation: L10–L11.

Apply the gradient to the opening loss

Training is gradient descent on the landscape

$\nabla\mathcal{L}$ points straight **uphill** — so training repeatedly steps **against** it:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\,\nabla\mathcal{L}(\boldsymbol{\theta}_t)$$

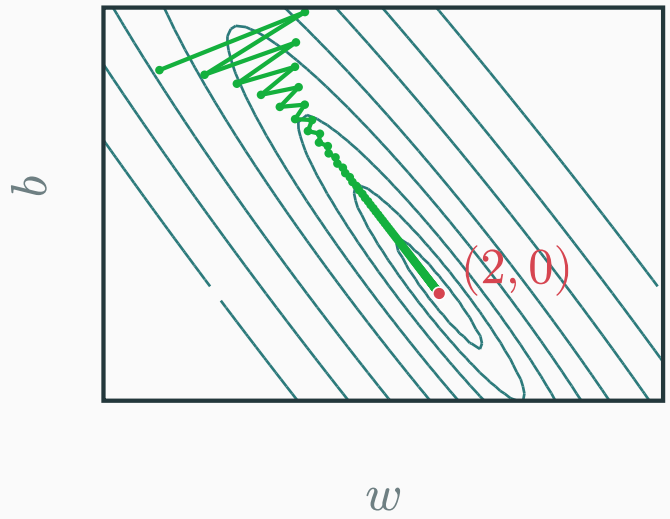
- Cross contour lines perpendicularly, downhill, at speed $\propto$ steepness.
- η (the **learning rate**) scales the step; L16–L18 derive its stable range and connect it to curvature.

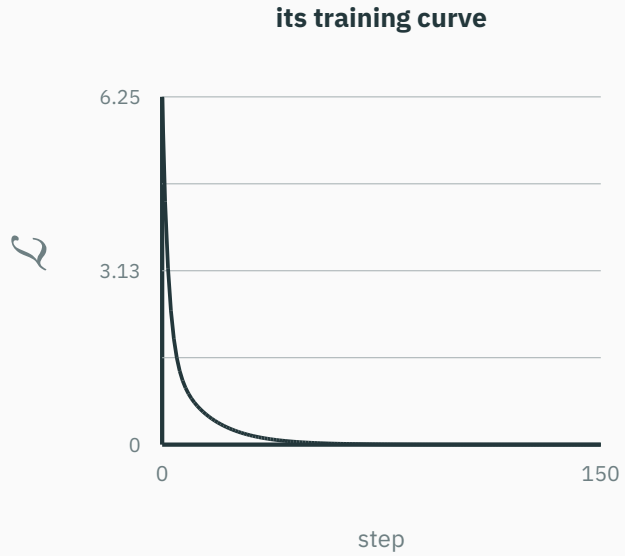
Watch it run on the loss **you built by hand** — next slide.

The walk and its altimeter

V VISUAL

Gradient descent on $\mathcal{L}(w, b)$ from the two-point fit, starting at $(w, b) = (-0.5, 2.5)$:





For this quadratic loss, the fast drop is followed by a long crawl **along the valley floor** toward $(2, 0)$. Real training curves can flatten for several other reasons, so the curve alone does not diagnose a valley.

Interactive: gradient-descent trajectories

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

Compare gradient descent and other optimizers on 2-D losses. Relate the contour spacing, gradient direction, and trajectory to local curvature.

Try the ill-conditioned valley: watch steps zigzag **across** the valley precisely because each one is perpendicular to its contour.

Where today's gradient is used later

lecture	what it does with today's gradient
L9	derivatives become **matrices**: Jacobian (many outputs), Hessian (curvature)
L10–L11	how ∇f is actually **computed**: autodiff, backprop, `loss.backward()`
L16–L18	descent **derived and tuned**: step sizes, convergence, Newton
L19–L20	constrained optima: where the constraint's and the loss's gradients **align**

Every one of those lectures manipulates the object built today: the vector of all partial sensitivities.

Summary & what’s next

Lecture 8 — summary

- **Two views**: surface, and contour map of level sets $f = K$; crowding = steepness.
- **Partials**: freeze all but one variable, differentiate as in L7 — ratios 4 and 6 at $(2, 1)$.
- **Gradient**: $\nabla f = (\partial f/\partial x, \partial f/\partial y)$ — one arrow per point, on the **map**.
- **Directional derivative** ⭐: $D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u}$ — one vector, every slope.
- **Geometry** ⭐: $\nabla f \perp$ contours (level walk $\Rightarrow$ dot $= 0$); steepest ascent ∇f at rate $\|\nabla f\|$, steepest descent $-\nabla f$ — Cauchy–Schwarz, from L3.
- **Chain rule with branches**: multiply along routes, **add** the routes — backprop's skeleton.
- **Training**: each update follows a local descent direction, and the training curve records the objective value along that trajectory.

Lecture 8 — summary

Read before L9: MML §5.2–§5.3 (partial differentiation & gradients). **T4** this week: `meshgrid`, contours, quivers, and your own two-parameter descent loop in NumPy.

Practice problems

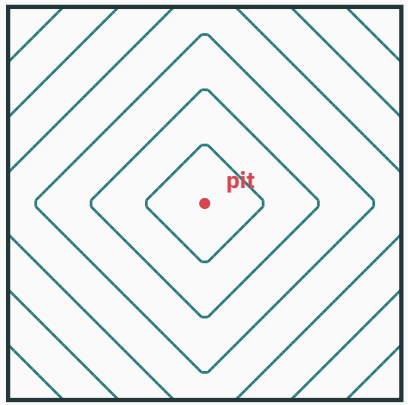
Try on paper; verify in the T4 notebook.

1. For $f(x, y) = x^3 y + y^2$: compute both partials at $(1, 2)$ and assemble ∇f.
2. Sketch the contour map of $f(x, y) = 2x + y$. What shapes are the level sets? Compute ∇f — why is it the **same arrow everywhere**, and how do the contours confirm it?
3. On our bowl at $(1, 1)$: find both unit directions with **zero** instantaneous change, and verify each with a dot product.
4. For $f = xy$ at $(2, 3)$: the slope along $\boldsymbol{u} = (1, 1)/\sqrt{2}$, and the steepest slope available there.
5. Let $x = \cos t$, $y = \sin t$ on the bowl $f = x^2 + 3y^2$. Compute $\mathrm{d}f/\,\mathrm{d}t$ at $t = \pi/4$ twice: by the branching chain rule, and by substituting first.
6. A quiver plot shows every arrow pointing **toward** the origin, shrinking near it. Sketch plausible contours, and propose an f that fits.

⭐⭐⭐ Where the map has corners

★ OPTIONAL

$f(x, y) = |x| + |y|$ — the L1 norm from L3, as terrain:



Diamond rings — with **corners** wherever $x = 0$ or $y = 0$.

At a corner the left and right slices disagree on the slope: **no single tangent, no gradient**. The smooth local-plane argument no longer applies.

At a nondifferentiable corner, a **subgradient** replaces the derivative. This is used for L_1 regularization and ReLU networks; L16 revisits the convex case.

The gradient points where the contour lines crowd.

Next: **Jacobian, Hessian & Multivariate Taylor** — when outputs are vectors too, the derivative becomes a matrix.